

THE ECONOMIC PROBLEM

How would you choose to use or allocate...

- 6 hours to study for Math, English and Econ exams?
- 10 pounds of your monthly allowance?
- 5 feddans of agricultural land?
- 50 million pounds if you were the Governor of Khartoum?

Scarcity of Resources* → *Economic Problem* → *Question of Choice

The Basic Economic Questions:

- **What** to produce?
- **How much** to produce?
- **How** to produce?
- **For whom** to produce?

(Cont.)

(a) What to produce?

- Do we produce everything in this country? If not, which goods will we produce here and which goods will we import?

(b) How much to produce?

- Once we have decided what goods to produce, next question: How much of each good will we produce?

(c) How to produce?

- Suppose we have decided on quantities of goods that will be produced, next question: What techniques to use in production?

(d) For whom to produce?

- Who benefits from the goods that were produced?

Technological Choices - The **Production Possibilities Frontier** (PPF):

- PPF $\Rightarrow$ a tool useful for illustrating choices available to society and its constraints.
- PPF $\equiv$ all the possible combinations of the **maximum** amounts of two goods/services that can be produced with a given amount of resources.

Assumptions about the PPF:

- Only **two goods** are produced: food (F) and clothing (C).
- The society has **limited** resources (in fixed supply).
- Resources available are land, labor, and capital
- Capital $\equiv$ physical capital like machinery, buildings, etc.

Example of PPF: all resources used

Trade-off

OC or rate
of trade-off

	FOOD (F)	CLOTHING (C)	ΔF	ΔC	$\frac{\Delta C}{\Delta F}$
A	0	1000			
B	100	950	100	-50	-0.5
C	200	850	100	-100	-1
D	300	650	100	-200	-2.0
E	400	400	100	-250	-2.5
F	500	0	100	-400	-4

Specializing in
clothing prodn

Speciali
-zing
in food
prodn

Notes on the PPF:

(1) Interpretation of points, say point D:

- If we use all resources, we can produce at most 300 units of Food and 650 units of Clothing

(2) Trade-off:

- If we want to ↑ Food production, must ↓ Clothing

(3) Opportunity Cost of Food:

- amount of Clothing that must be given up to produce an extra unit of Food.
- Between points A and B, OC of F is 0.5.

(Cont.)

(4) **Opportunity cost:**

- the value of the best foregone alternative.

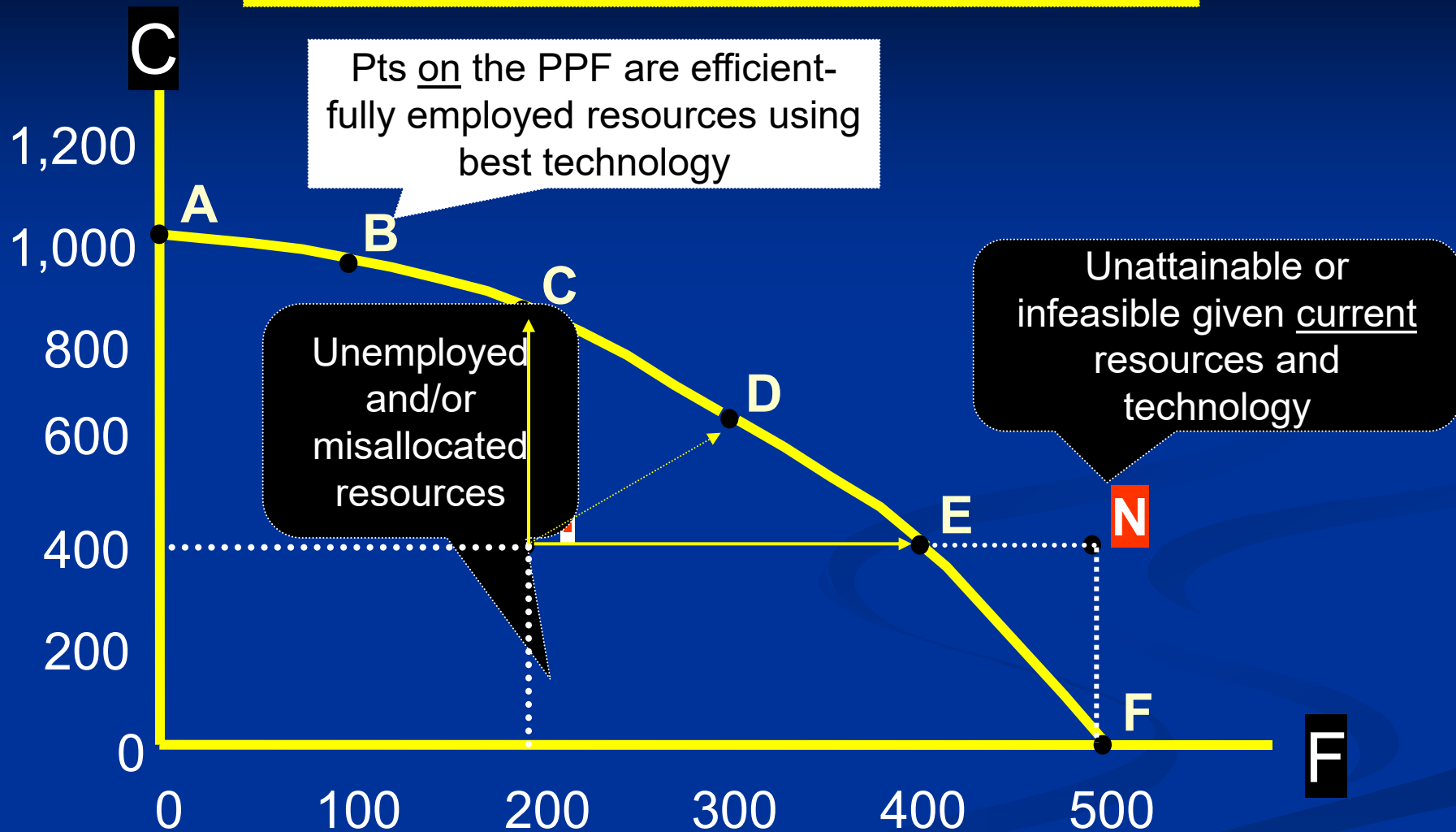
(5) Rate of trade-off is increasing:

- in absolute value $\Rightarrow$ increasing opportunity cost of F

(6) **Implication:**

- As we produce more of F, it gets harder to substitute F for C

Graphical Illustration of PPF



Does the PPF Answer the Basic Economic Questions?

■ What to produce?

- Shown by 2 goods on axes of PPF.

■ How much to produce?

- Shown by points along PPF.

■ How to produce?

- Use technology that will enable economy to produce goods represented by points along PPF.

■ For whom to produce?

- Not shown by PPF.

Shifts in PPF

